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2027_IV_CSBS_Machine Learning Lab

PSG iTech_2027_Machine Learning 6_COD

Attempt : 1
Total Mark : 20
Marks Obtained : 20

Section 1 : Coding

1. Problem Statement

Daniel is developing a Bayesian Network model to predict the probability of heart disease based on specific attributes like age, gender, family history, diet, lifestyle, and cholesterol levels. Given a CSV dataset of historical patient records, write a program that loads the dataset, constructs a probability table, implements the Bayesian Network using dictionaries, and performs inference to determine the probability of heart disease given user-provided evidence.

Answer

main.py

```
import os  
import pandas as pd
```

```

file_name = input().strip()
current_directory = os.path.dirname(__file__)
file_path = os.path.join(current_directory, file_name)

df = pd.read_csv(file_path)

def create_probability_table(df):
    total_count = len(df)
    if total_count == 0:
        return None, "Dataset is empty."

    hd_yes = df[df['heartdisease'] == 1]
    hd_no = df[df['heartdisease'] == 0]

    if len(hd_yes) == 0 or len(hd_no) == 0:
        return None, "Not enough data for prediction."

    p_hd_yes = len(hd_yes) / total_count
    p_hd_no = len(hd_no) / total_count

    probability_table = {"P(heartdisease=1)": p_hd_yes, "P(heartdisease=0)":
p_hd_no}
    alpha = 1

    for feature in df.columns:
        if feature == "heartdisease":
            continue
        probability_table[feature] = {}
        unique_values = df[feature].nunique()
        for value in df[feature].unique():
            p_feature_given_yes = (len(hd_yes[hd_yes[feature] == value]) + alpha) /
(len(hd_yes) + unique_values * alpha)
            p_feature_given_no = (len(hd_no[hd_no[feature] == value]) + alpha) /
(len(hd_no) + unique_values * alpha)
            probability_table[feature][value] = {
                "P(value|heartdisease=1)": p_feature_given_yes,
                "P(value|heartdisease=0)": p_feature_given_no
            }

    return probability_table, None

def infer_heart_disease(probability_table, evidence):

```

```
if probability_table is None:  
    return "Error: Probability table not available."
```

```
p_hd_yes = probability_table["P(heartdisease=1)"]  
p_hd_no = probability_table["P(heartdisease=0)"]
```

```
for feature, value in evidence.items():  
    if feature not in probability_table or value not in probability_table[feature]:  
        return f"Feature '{feature}' not found in dataset."  
    p_hd_yes *= probability_table[feature][value]["P(value|heartdisease=1)"]  
    p_hd_no *= probability_table[feature][value]["P(value|heartdisease=0)"]
```

```
total = p_hd_yes + p_hd_no  
if total == 0:  
    return "Error: Not enough data for prediction."
```

```
p_hd_yes /= total  
p_hd_no /= total
```

```
return {"Heart Disease (Yes)": p_hd_yes, "Heart Disease (No)": p_hd_no}
```

```
evidence = {  
    "age": int(input().strip()),  
    "Gender": int(input().strip()),  
    "Family": int(input().strip()),  
    "diet": int(input().strip()),  
    "Lifestyle": int(input().strip()),  
    "cholesterol": int(input().strip()),  
}
```

```
probability_table, error_message = create_probability_table(df)
```

```
if error_message:  
    pass  
else:  
    result = infer_heart_disease(probability_table, evidence)  
    if isinstance(result, str):  
        pass  
    else:  
        print("Predicted Probabilities:")  
        for key, value in result.items():  
            print(f"{key}: {value:.4f}")
```

data.csv

age,Gender,Family,diet,Lifestyle,cholesterol,heartdisease

0,0,1,1,3,0,1
0,1,1,1,3,0,1
1,0,0,0,2,1,1
4,0,1,1,3,2,0
3,1,1,0,0,2,0
2,0,1,1,1,0,1
4,0,1,0,2,0,1
0,0,1,1,3,0,1
3,1,1,0,0,2,0
1,1,0,0,0,2,1
4,1,0,1,2,0,1
4,0,1,1,3,2,0
2,1,0,0,0,0,0
2,0,1,1,1,0,1
3,1,1,0,0,1,0
0,0,1,0,0,2,1
1,1,0,1,2,1,1
3,1,1,1,0,1,0
4,0,1,1,3,2,0

data1.csv

age,Gender,Family,diet,Lifestyle,cholesterol,heartdisease

1,0,1,0,2,1,1
3,1,1,1,3,2,0
0,0,0,1,0,1,1
2,1,0,0,1,0,0
4,0,1,1,2,0,1
1,1,0,1,3,1,1
2,0,1,0,1,2,0
0,1,1,1,3,0,1
3,0,1,0,2,1,1
4,1,0,1,2,2,0

data2.csv

age,Gender,Family,diet,Lifestyle,cholesterol,heartdisease

0,1,1,1,3,0,1
2,0,1,0,1,1,0
3,1,0,1,2,2,1
1,0,1,0,0,0,1
4,0,0,1,3,1,0

0,1,1,1,2,0,1
3,1,0,0,0,2,0
1,1,1,0,1,0,1
4,1,1,1,3,2,0
2,0,0,1,1,1,1
0,1,1,1,2,0,1
3,0,1,0,1,2,0
4,1,0,1,2,1,0
1,0,1,1,3,0,1
2,1,0,0,0,2,1

data3.csv

age,Gender,Family,diet,Lifestyle,cholesterol,heartdisease

0,0,1,1,3,0,1
1,1,1,1,2,1,0
2,0,0,0,1,2,1
3,1,1,1,3,2,0
4,0,1,0,2,0,1
0,1,0,1,1,0,1
3,0,1,0,0,2,0
1,1,0,1,2,1,1
4,0,0,1,3,2,0
2,1,1,0,0,0,1
0,0,0,1,1,1,0
3,1,1,0,0,2,1
4,1,0,1,2,0,1
2,0,1,1,1,2,0
0,1,1,0,3,1,1
1,0,0,1,2,1,1
4,1,1,0,2,0,0
3,0,0,1,1,1,1
1,1,1,0,3,2,0
2,0,0,1,1,1,1
0,1,1,1,3,0,1
4,0,0,0,2,1,1
1,1,1,1,3,2,0
3,0,1,0,2,1,1
2,1,0,1,0,0,1

data4.csv

age,Gender,Family,diet,Lifestyle,cholesterol,heartdisease

3,1,0,1,2,2,0

1,0,1,1,1,0,1
4,0,0,1,3,1,0
2,1,1,0,0,2,1
0,1,0,1,1,1,1
3,0,1,0,2,1,0
1,1,0,1,3,0,1
2,0,1,0,0,2,0

data5.csv

age,Gender,Family,diet,Lifestyle,cholesterol,heartdisease

0,0,1,1,3,0,1
1,1,1,1,2,1,0
2,0,0,0,1,2,1
3,1,1,1,3,2,0
4,0,1,0,2,0,1
0,1,0,1,1,0,1
3,0,1,0,0,2,0
1,1,0,1,2,1,1
4,0,0,1,3,2,0
2,1,1,0,0,0,1
0,0,0,1,1,1,0
3,1,1,0,0,2,1
4,1,0,1,2,0,1
2,0,1,1,1,2,0
0,1,1,0,3,1,1
1,0,0,1,2,1,1
4,1,1,0,2,0,0
3,0,0,1,1,1,1
1,1,1,0,3,2,0
2,0,0,1,1,1,1
0,1,1,1,3,0,1
4,0,0,0,2,1,1
1,1,1,1,3,2,0
3,0,1,0,2,1,1
2,1,0,1,0,0,1
3,0,1,1,2,1,0
1,0,0,0,3,2,1
4,1,1,1,2,0,0
0,1,1,0,1,1,1
2,0,0,1,3,2,1

data6.csv

age,Gender,Family,diet,Lifestyle,cholesterol,heartdisease

1,0,1,1,2,1,1

3,1,1,0,0,2,0

0,1,1,1,3,0,1

2,0,1,0,1,1,0

4,1,0,1,2,0,1

data7.csv

age,Gender,Family,diet,Lifestyle,cholesterol,heartdisease

1,0,1,1,2,1,1

3,1,1,1,3,2,0

0,0,0,1,0,1,1

2,1,0,0,1,0,1

4,0,1,1,2,0,1

1,1,0,1,3,1,1

2,0,1,0,1,2,1

0,1,1,1,3,0,1

3,0,1,0,2,1,1

4,1,0,1,2,2,1

2,1,0,1,0,1,1

4,0,1,1,3,2,1

1,0,0,0,2,1,1

3,1,1,1,0,1,0

Status : Correct

Marks : 10/10

2. Problem Statement

Liam, a healthcare data analyst, needs to analyze a dataset containing COVID-19 diagnoses and associated symptoms. Write a program to read a dataset file and construct probability tables using Bayesian inference. Based on a user's symptom inputs, calculate and display the probability of having COVID-19.

Answer

main.py

```
import os
import pandas as pd
```

```

def create_probability_table(df):
    total_count = len(df)
    if total_count == 0:
        return None, "Dataset is empty."

    covid_yes = df[df['Diagnosis'] == 1]
    covid_no = df[df['Diagnosis'] == 0]

    if len(covid_yes) == 0 or len(covid_no) == 0:
        return None, "Not enough data for prediction."

    p_covid_yes = len(covid_yes) / total_count
    p_covid_no = len(covid_no) / total_count

    probability_table = {
        "P(Diagnosis=1)": p_covid_yes,
        "P(Diagnosis=0)": p_covid_no
    }

    alpha = 1

    for feature in df.columns:
        if feature == "Diagnosis":
            continue
        probability_table[feature] = {}
        unique_values = df[feature].nunique()
        for value in df[feature].unique():
            p_feature_given_yes = (len(covid_yes[covid_yes[feature] == value]) +
alpha) / (len(covid_yes) + unique_values * alpha)
            p_feature_given_no = (len(covid_no[covid_no[feature] == value]) + alpha) /
(len(covid_no) + unique_values * alpha)
            probability_table[feature][value] = {
                "P(value|Diagnosis=1)": p_feature_given_yes,
                "P(value|Diagnosis=0)": p_feature_given_no
            }

    return probability_table, None

def infer_covid_diagnosis(probability_table, evidence):
    if probability_table is None:
        return "Error: Probability table not available."

```



```
p_covid_yes = probability_table["P(Diagnosis=1)"]
p_covid_no = probability_table["P(Diagnosis=0)"]
```

```
for feature, value in evidence.items():
    if feature not in probability_table or value not in probability_table[feature]:
        return f"Feature '{feature}' not found in dataset."
```

```
p_covid_yes *= probability_table[feature][value]["P(value|Diagnosis=1)"]
p_covid_no *= probability_table[feature][value]["P(value|Diagnosis=0)"]
```

```
total = p_covid_yes + p_covid_no
if total == 0:
    return "Error: Not enough data for prediction."
```

```
p_covid_yes /= total
p_covid_no /= total
```

```
return {
    "COVID-19 Positive": p_covid_yes,
    "COVID-19 Negative": p_covid_no
}
```

```
if __name__ == "__main__":
    file_name = input().strip()
    current_directory = os.path.dirname(__file__)
    file_path = os.path.join(current_directory, file_name)
```

```
try:
    df = pd.read_csv(file_path)
except Exception as e:
    exit()
```

```
evidence = {
    "Fever": int(input().strip()),
    "Cough": int(input().strip()),
    "Fatigue": int(input().strip()),
    "Loss of Smell": int(input().strip()),
    "Hypertension": int(input().strip()),
    "Diabetes": int(input().strip()),
    "Respiratory Disease": int(input().strip()),
    "Contact with Infected": int(input().strip()),
    "Travel History": int(input().strip())
}
```

```
}
```

```
probability_table, error_message = create_probability_table(df)
```

```
if error_message:
```

```
    exit()
```

```
else:
```

```
    result = infer_covid_diagnosis(probability_table, evidence)
```

```
    if isinstance(result, dict):
```

```
        print("Predicted Probabilities:")
```

```
        print(f"COVID-19 Positive: {result['COVID-19 Positive']:.4f}")
```

```
        print(f"COVID-19 Negative: {result['COVID-19 Negative']:.4f}")
```

```
    else:
```

```
        exit()
```

data.csv

Fever,Cough,Fatigue,Loss of Smell,Hypertension,Diabetes,Respiratory
Disease,Contact with Infected,Travel History,Diagnosis

1,1,0,1,0,0,0,1,0,1

0,1,1,0,1,0,1,0,1,0

1,0,1,1,0,1,0,0,0,1

0,0,0,0,0,0,0,0,0,0

1,1,1,1,1,1,1,1,1,1

0,0,1,0,0,0,0,1,0,0

1,0,0,1,0,0,1,1,1,1

0,1,0,0,1,0,0,0,1,0

1,1,1,1,1,1,0,1,0,1

0,0,0,0,0,0,0,0,0,0

data1.csv

Fever,Cough,Fatigue,Loss of Smell,Hypertension,Diabetes,Respiratory
Disease,Contact with Infected,Travel History,Diagnosis

1,1,0,1,0,0,0,1,0,1

0,1,1,0,1,0,1,0,1,0

1,0,1,1,0,1,0,0,0,1

0,0,0,0,0,0,0,0,0,0

1,1,1,1,1,1,1,1,1,1

0,0,1,0,0,0,0,1,0,0

1,0,0,1,0,0,1,1,1,1

0,1,0,0,1,0,0,0,1,0

1,1,1,1,1,1,0,1,0,1

0,0,0,0,0,0,0,0,0,0

1,1,0,0,0,1,1,1,0,1
0,0,1,1,0,0,0,1,1,0
1,0,1,0,1,1,1,0,1,1
0,1,0,1,0,0,1,0,0,0
1,0,0,0,1,0,1,1,1,1
0,1,1,1,0,1,0,0,0,0
1,1,0,1,1,1,0,1,0,1
0,0,1,0,0,0,1,0,1,0
1,1,1,0,1,1,0,1,1,1
0,0,0,1,0,0,0,0,0,0

data2.csv

Fever,Cough,Fatigue,Loss of Smell,Hypertension,Diabetes,Respiratory
Disease,Contact with Infected,Travel History,Diagnosis

1,1,0,1,0,0,0,1,0,1
0,1,1,0,1,0,1,0,1,0
1,0,1,1,0,1,0,0,0,1
0,0,0,0,0,0,0,0,0,0
1,1,1,1,1,1,1,1,1,1
0,0,1,0,0,0,0,1,0,0
1,0,0,1,0,0,1,1,1,1
0,1,0,0,1,0,0,0,1,0
1,1,1,1,1,1,0,1,0,1
0,0,0,0,0,0,0,0,0,0
1,1,0,0,0,1,1,1,0,1
0,0,1,1,0,0,0,1,1,0
1,0,1,0,1,1,1,0,1,1
0,1,0,1,0,0,1,0,0,0
1,0,0,0,1,0,1,1,1,1
0,1,1,1,0,1,0,0,0,0
1,1,0,1,1,1,0,1,0,1
0,0,1,0,0,0,1,0,1,0
1,1,1,0,1,1,0,1,1,1
0,0,0,1,0,0,0,0,0,0
1,0,1,1,0,1,0,1,0,1
0,1,0,0,1,0,1,0,1,0
1,1,0,0,0,1,1,1,1,1
0,0,1,1,1,0,0,0,0,0
1,0,0,1,1,1,1,1,0,1
0,1,1,0,0,0,0,1,1,0
1,1,1,1,1,0,1,0,1,1
0,0,0,0,0,1,0,0,0,0

1,0,1,0,1,1,1,1,1,1
0,1,0,1,0,0,0,0,0,0

data3.csv

Fever,Cough,Fatigue,Loss of Smell,Hypertension,Diabetes,Respiratory
Disease,Contact with Infected,Travel History,Diagnosis

1,1,0,1,0,0,0,1,0,1
0,1,1,0,1,0,1,0,1,0
1,0,1,1,0,1,0,0,0,1
0,0,0,0,0,0,0,0,0,0
1,1,1,1,1,1,1,1,1,1
0,0,1,0,0,0,0,1,0,0
1,0,0,1,0,0,1,1,1,1
0,1,0,0,1,0,0,0,1,0
1,1,1,1,1,1,0,1,0,1
0,0,0,0,0,0,0,0,0,0
1,1,0,0,0,1,1,1,0,1
0,0,1,1,0,0,0,1,1,0
1,0,1,0,1,1,1,0,1,1
0,1,0,1,0,0,1,0,0,0
1,0,0,0,1,0,1,1,1,1
0,1,1,1,0,1,0,0,0,0
1,1,0,1,1,1,0,1,0,1
0,0,1,0,0,0,1,0,1,0
1,1,1,0,1,1,0,1,1,1
0,0,0,1,0,0,0,0,0,0
1,0,1,1,0,1,0,1,0,1
0,1,0,0,1,0,1,0,1,0
1,1,0,0,0,1,1,1,1,1
0,0,1,1,1,0,0,0,0,0
1,0,0,1,1,1,1,1,0,1
0,1,1,0,0,0,0,1,1,0
1,1,1,1,1,0,1,0,1,1
0,0,0,0,0,1,0,0,0,0
1,0,1,0,1,1,1,1,1,1
0,1,0,1,0,0,0,0,0,0
1,1,0,1,1,0,1,1,0,1
0,0,1,0,1,1,0,0,1,0
1,0,1,1,0,1,1,1,1,1
0,1,0,0,0,0,0,0,0,0
1,1,1,0,1,0,1,1,0,1
0,0,0,1,0,1,0,0,1,0

1,0,0,1,1,1,1,1,1,1
0,1,1,0,0,0,0,1,0,0
1,1,0,0,1,1,0,1,1,1
0,0,1,1,0,0,1,0,0,0
1,0,1,0,1,1,1,1,0,1
0,1,0,1,0,0,0,0,1,0
1,1,1,1,1,1,0,1,1,1
0,0,0,0,0,0,0,0,0,0
1,1,0,0,0,1,1,1,0,1
0,0,1,1,0,0,0,1,1,0
1,0,1,0,1,1,1,0,1,1
0,1,0,1,0,0,1,0,0,0
1,0,0,0,1,0,1,1,1,1
0,1,1,1,0,1,0,0,0,0

data4.csv

Fever,Cough,Fatigue,Loss of Smell,Hypertension,Diabetes,Respiratory
Disease,Contact with Infected,Travel History,Diagnosis

1,1,0,1,0,0,0,1,0,1
0,1,1,0,1,0,1,0,1,0
1,0,1,1,0,1,0,0,0,1
0,0,0,0,0,0,0,0,0,0
1,1,1,1,1,1,1,1,1,1
0,0,1,0,0,0,0,1,0,0
1,0,0,1,0,0,1,1,1,1
0,1,0,0,1,0,0,0,1,0
1,1,1,1,1,1,0,1,0,1
0,0,0,0,0,0,0,0,0,0
1,1,0,1,0,1,1,1,0,1
0,1,1,0,1,0,0,1,1,0
1,0,1,1,0,1,0,0,1,1
0,0,0,0,0,0,1,0,0,0
1,1,1,1,1,1,1,1,1,1
0,1,0,0,1,0,0,0,1,0
1,0,0,1,0,0,1,1,1,1
0,1,1,1,0,1,0,0,0,0
1,1,1,0,1,1,0,1,1,1
0,0,0,1,0,0,0,0,0,0
1,1,0,1,0,1,1,1,0,1
0,1,1,0,1,0,0,1,1,0
1,0,1,1,0,1,0,0,1,1
0,0,0,0,0,0,1,0,0,0

1,1,1,1,1,1,1,1,1,1
0,1,0,0,1,0,0,0,1,0
1,0,0,1,0,0,1,1,1,1
0,1,1,1,0,1,0,0,0,0
1,1,1,0,1,1,0,1,1,1
0,0,0,1,0,0,0,0,0,0
1,1,0,1,0,1,1,1,0,1
0,1,1,0,1,0,0,1,1,0
1,0,1,1,0,1,0,0,1,1
0,0,0,0,0,0,1,0,0,0
1,1,1,1,1,1,1,1,1,1
0,1,0,0,1,0,0,0,1,0
1,0,0,1,0,0,1,1,1,1
0,1,1,1,0,1,0,0,0,0
1,1,1,0,1,1,0,1,1,1
0,0,0,1,0,0,0,0,0,0
1,1,0,1,0,1,1,1,0,1
0,1,1,0,1,0,0,1,1,0
1,0,1,1,0,1,0,0,1,1
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1,1,1,1,1,1,1,1,1,1
0,1,0,0,1,0,0,0,1,0
1,0,0,1,0,0,1,1,1,1
0,1,1,1,0,1,0,0,0,0
1,1,1,0,1,1,0,1,1,1
0,0,0,1,0,0,0,0,0,0

data5.csv

Fever,Cough,Fatigue,Loss of Smell,Hypertension,Diabetes,Respiratory
Disease,Contact with Infected,Travel History,Diagnosis

1,1,0,1,0,0,0,1,0,1
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1,1,1,1,1,1,0,1,0,1
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0,1,1,0,1,0,0,1,1,0

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1,1,1,1,1,1,1,1,1,1
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1,0,0,1,0,0,1,1,1,1
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1,1,1,0,1,1,0,1,1,1
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0,1,1,0,1,0,0,1,1,0
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1,1,1,1,1,1,1,1,1,1
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0,0,0,1,0,0,0,0,0,0
1,0,1,0,1,1,1,1,1,1
0,1,0,1,0,0,0,0,0,0

data6.csv

Fever,Cough,Fatigue,Loss of Smell,Hypertension,Diabetes,Respiratory
Disease,Contact with Infected,Travel History,Diagnosis

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0,1,0,0,1,0,1,0,1,0

data7.csv

Fever,Cough,Fatigue,Loss of Smell,Hypertension,Diabetes,Respiratory
Disease,Contact with Infected,Travel History,Diagnosis

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Status : Correct

Marks : 10/10